Speed

Exam code: 0607

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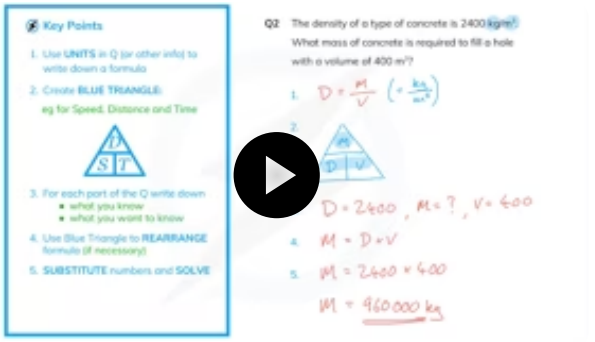

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Speed

What is speed?

- **Speed** is equal to **distance** divided by **time**
- **Speed** is commonly measured in **metres per second** (**m/s**) or **kilometres per hour** (**km/h**)
  - The units indicate **speed** is **distance per time**

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

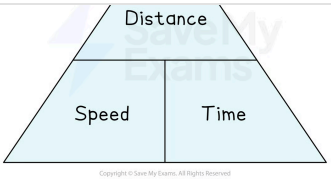
- '**Speed**' (in this formula) means '**average speed**'
- The formula can be rearranged to
  - $\text{Time} = \frac{\text{Distance}}{\text{Speed}}$
  - $\text{Distance} = \text{Speed} \times \text{Time}$
- In harder problems there are often **two journeys** or **two parts to one longer journey**

What is a formula triangle?

- A **formula triangle** shows the **relationship** between the **different measures** in a **compound formula**
- They can be useful if you are less confident with rearranging formulas
- For Speed, Distance and Time:

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



- If you are calculating a variable on the **top of the triangle**, multiply the two variables on the bottom
  - For example, $\text{Distance} = \text{Speed} \times \text{Time}$
- If you are calculating a variable on the **bottom of the triangle**, divide the top by the other variable on the bottom
  - For example, $\text{Speed} = \text{Distance} \div \text{Time}$ and $\text{Time} = \text{Distance} \div \text{Speed}$

Examiner Tips and Tricks

- Look out for a mixture of units
  - **Time** can be given as **minutes** but common phrases like '**half an hour**' (30 minutes) could also be used in the same question
  - Any mixed units should be those in common use and easy to convert, e.g. **m** to **km** etc

Worked Example

(a) A runner completes a 10 km race in three quarters of an hour.

Find the average speed of the runner in metres per second. Round your answer to 3 significant figures.

As the answer must be in metres per second, convert the distance to metres, and the time to seconds

Multiply by 1000 to convert km to metres

10 km = 10 000 m

Write the time in minutes, you could calculate this as $\frac{3}{4} \times 60 \text{ mins}$

Three quarters of an hour = 0.75 hrs = 45 mins

Multiply by 60 to convert minutes to seconds

45 × 60 = 2 700 seconds

Use the formula $\text{Speed} = \frac{\text{Distance}}{\text{Time}}$

Round to 3 significant figures

3.70 m/s

(b) A sprinter's average speed when running 100 m is 8.85 m/s.

Calculate the total time taken in seconds for the sprinter to run 100 m.
Round your answer to the nearest hundredth of a second.

The units are already in metres, and metres per second, so do not need converting to reach an answer in seconds

Use the formula $\text{Speed} = \frac{\text{Distance}}{\text{Time}}$

$$8.85\ \text{m/s} = \frac{100\ \text{m}}{\text{Time}}$$

Rearrange to make the time the subject
Multiply both sides by time, and then divide both sides by the speed

$$8.85 \times \text{Time} = 100$$
$$\text{Time} = \frac{100}{8.85}$$
$$\text{Time} = 11.29943503...$$

Round to the nearest hundredth of a second (2 decimal places)

11.30 s

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